Giovanni Guidarini

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Education

MSc in Computational Science and Engineering Sep. 2025 – Present
ETH Zürich *Zürich, Switzerland*

- Expected graduation: february 2028. GPA: 5.78/6
- Current coursework: VLSI 1: HDL-Based Design for FPGAs; Embedded Systems.
- Relevant coursework: HPC Lab; Large-Scale Convex Optimization; AI in the Sciences and Engineering; Programming Techniques; Probabilistic Artificial Intelligence.

BSc in Mathematical Engineering Sep. 2022 – Jul. 2025
Politecnico di Milano *Milan, Italy*

- Graduated 110/110 with Honors
- Relevant coursework: Linear Optimization; Numerical Analysis; Linear Algebra; Probability Theory; PDEs.

Projects

FPGA Numerical Accelerator for 2D Poisson Problems | *SystemVerilog, FPGA, Fixed-Point* Ongoing

- Designing a fixed-point FPGA accelerator for the conjugate-gradient method applied to the 2D Poisson equation; implemented streaming vector arithmetic engines, a diffusion stencil, and a dot-product reduction in SystemVerilog.
- Verified all implemented RTL modules bit-for-bit against a fixed-point C++ reference using self-checking testbenches.
- Developing memory routing and control logic, with planned synthesis and deployment on a Zybo Z7 FPGA board, followed by timing, latency, and hardware resource optimization.

Particle Track Reconstruction as a Spin-Glass Optimization Problem | *Optimization, C++, Python*
ETH Zürich, Computational Statistical Physics Spring 2026

- Mapped LHC particle-track reconstruction to an Ising-like spin-glass optimization problem and built a C++ simulated-annealing solver with a physics-motivated Hamiltonian encoding reconstructed tracks' geometric constraints
- Tuned solver hyperparameters with Optuna on the Euler cluster, reaching 85% strict track efficiency on clean events; developed a reproducible computational workflow for evaluating optimization choices.

Generative Antibody Design with Protein Language Models | *Deep Learning, PyTorch, Optimization*
ETH Zürich, Learning & Adaptive Systems Group Spring 2026

- Implemented beam search over the CDRH3 region to use ESM2 as a generative model for antibody design; fine-tuned it with evolutionary fine-tuning and Direct Preference Optimization on experimental binding-affinity data.
- Built an end-to-end training and evaluation pipeline with Hydra configurations and Slurm cluster jobs, comparing vanilla, evotuned, and DPO-tuned models across correlation, diversity, affinity, and out-of-distribution generalization.

AI in the Sciences and Engineering | *Deep Learning, PyTorch, Optimization* Fall 2025

- Analyzed optimization pathologies in Physics-Informed Neural Networks on a multiscale Poisson equation using filter normalization and Hessian-eigenvector analysis, identifying high-frequency components as a source of ill-conditioned loss landscapes and training instability.
- Trained a Fourier Neural Operator surrogate for an unknown dynamical system; implemented alternative training strategies, evaluated resolution invariance, and demonstrated transfer learning to out-of-distribution regimes.

Experience

Artificial Intelligence team intern Jul. 2024
Anima SGR *Milan, Italy*

- Independently built and backtested a data-to-results pipeline for a sentiment-driven sector-rotation strategy on STOXX 600 equities (2012–2024); evaluated portfolio-construction and concentration choices, achieving 0.50 Sharpe vs. 0.31 benchmark and 4.9% annualized excess returns.
- Presented the findings to the Head of the Quantitative Division and CFO; the strategy was subsequently adopted into the firm's production pipeline.

Technical Skills

Hardware / systems: SystemVerilog, FPGA, fixed-point arithmetic
Programming: C/C++, Python, CUDA, PyTorch, MPI, OpenMP, Git, Linux, Slurm

Additional Information

Languages: Italian (Native), English (IELTS 8 - C1)
Interests: Motorsports, AI, Art, History and Literature, former competitive rugby player